Searching for Emergent Geometric Coordinate Systems in LLM Hidden Representations: A Replication Attempt

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Abstract

We attempt to reconstruct a continuous geometric coordinate system from the hidden-layer activations of two open-weight language models (Qwen2.5-0.5B and Mistral-7B-v0.1) by treating a probe set of relational concepts as nodes in a graph, computing a cosine-similarity kernel between their activation vectors, and diagonalizing the resulting graph Laplacian. We report six results. First: probe tokens drawn from calendar/cyclic categories (days of the week, months) form a distinct cluster separated from quantity/space/relation concepts, and this separation sharpens with model scale, exceeding a randomized-token control on every measure we tried; scaling the probe set from 77 to 500 tokens across 25 categories confirms this and refines it, the categories that separate cleanly turn out to be small, closed, enumerable named sets in general (also continents, zodiac signs, playing-card ranks, planets), not specifically cyclic or temporal ones, while categories chosen to have no expected structure collapse into one undifferentiated region. Second: neither the generic graph-Laplacian method nor a template-based, in-context-arithmetic PCA method reproduces the clean circular manifolds reported for cyclic concepts in prior work (Engels et al., 2024); a first attempt at causal activation patching also produced no effect whatsoever, for a specific, diagnosable methodological reason (patching only the final token position, which the model could route around). Third, after correcting that flaw: patching the subject-token position itself (standard causal-tracing methodology) produces a clean, strongly positive result: swapping in a donor day/month's representation flips the model's downstream answer to match the donor, with 83–100% accuracy through the network's early-to-mid layers, collapsing sharply to 0% past a well-defined depth (layer ∼20 of 32 for days, ∼17 of 32 for months), while a random-direction control of matched norm stays at exactly 0% at every single layer. Fourth, testing the stronger geometric claim directly: at the same subject-token position, months trace the cleanest 2D loop we found anywhere in this project (a near-convex, non-self-crossing polygon in exact calendar order), yet synthetically rotating a base month's own vector within that discovered 2D plane by the angle matching a real calendar offset produces the correct answer 0/60 times, statistically indistinguishable from rotating by a deliberately wrong, off-lattice angle (also 0/60); the top two principal components at that position explain under 40% of the variance, so a clean-looking 2D loop is not the same as a causally sufficient 2D code. Fifth, and most decisively: refitting the same idea in a full-rank subspace (using $n-1$ principal components, which capture essentially 100% of the variance among the n items' own vectors) and replacing the ad hoc per-pair rotation with a single best-fit orthogonal rotation generator R (via the orthogonal Procrustes problem, fit once across all consecutive pairs simultaneously) recovers the full causal effect: applying R^{offset} to a base item's own vector, with no real donor data involved at all, matches or slightly exceeds the real donor patch's accuracy (100% vs. 100% for days, 85% vs. 82% for months), while a matched fractional-power rotation (an off-lattice control in this higher-dimensional setting) collapses to 8–14%, near chance. Day/month identity at this position is genuinely organized as a single consistent rotation acting on a high-dimensional subspace, just not the naive 2D circle a first PCA plot suggests. Sixth, testing whether this mechanism generalizes to the other small closed sets surfaced by the 500-token scale-up (zodiac signs, planets, playing-card ranks): it partially does, playing-card ranks reach 82%/73% (real donor/full-rank rotation, both significant at $p < 10^{-6}$ vs. chance), zodiac signs reach a smaller but still statistically genuine 42%/33% ($p < 0.02$), while planets' 33%/17% is, with only 6 testable items, not statistically distinguishable from chance ($p = 0.17$). The three testable categories' accuracy tracks average subject-name token length in the direction the single-token-position leakage mechanism diagnosed in the first causal patching attempt predicts (1.00 for days/months, 1.15 for cards, 2.00–2.25 for planets/zodiac), though with confidence intervals this wide across only 3–5 data points, we treat this as consistent with that mechanism rather than as an independent confirmation of it. We report all six results, the bugs found and fixed along the way, and why we believe even the negative ones are informative rather than merely inconclusive.

1 Motivation

The starting hypothesis (recorded in the project's original planning notes, reproduced in Appendix A) was that a language model's hidden layer could be treated as "a discrete, non-spatial relational substrate," and that applying an exact-diagonalization / graph Laplacian pipeline to a structured set of probe tokens might cause a continuous, geometric concept-manifold to "spontaneously reconstruct itself" out of the raw activations (e.g., number words tracing a line, or cyclic concepts like days-of-week tracing a circle).

This is not a new technique: graph Laplacian eigenmaps are a standard, 20-year-old manifold-learning method [1]. But the underlying empirical question is current and real: recent work has shown that large language models represent some cyclic concepts (days of the week, months, years within a century) as literal circles in activation space, confirmed via causal intervention on Mistral-7B and Llama-3-8B [2]. Our goal was to see whether a simpler, more generic pipeline (bare-word probing plus a standard graph Laplacian, rather than in-context arithmetic prompts plus targeted causal patching) could reproduce that finding, or something like it, on hardware-constrained local models.

2 Method

2.1 Models

Two open-weight decoder-only models were used:

- **Qwen2.5-0.5B** (24 layers, hidden size 896), run in fp32 on a consumer GPU (RTX 3070 Ti, 8GB VRAM).
- **Mistral-7B-v0.1** (32 layers, hidden size 4096), run in 4-bit NF4 quantization (via the `bitsandbytes` library) to fit the same 8GB card, one of the model families in which circular representations were previously reported [2].

2.2 Probe set

A structured set of 77 probe tokens across 8 relational categories was built by hand: `numbers` (one–twenty), `days_of_week`, `months`, `directions` (north/south/up/down/left/right/…), `shapes` (circle, square, …), `causal_relations` (because, therefore, …), `spatial_relations` (above, below, …), and `comparison_relations` (more than, less than, …). This is smaller than the 500-token set originally envisioned, but large enough to test the pipeline honestly.

2.3 Phase 1: Activation extraction

A forward hook was registered on a mid-network decoder layer (layer 12 of 24 for Qwen; layer 16 of 32 for Mistral). For each probe token, we ran a forward pass and captured the hidden state at the *last* token position, not a mean pool over all subword tokens (see §3.1 for why this matters).

2.4 Phase 2: Relational graph construction

Activation vectors were standardized per-dimension (z-scored across the probe set) before computing pairwise cosine similarity, since a small number of outlier-magnitude dimensions are known to dominate raw cosine similarity in transformer hidden states. Negative similarities were clipped to zero (see §3.1), and the unnormalized graph Laplacian $L = D - W$ was assembled from the resulting non-negative weight matrix W.

2.5 Phase 3: Diagonalization

L was exactly diagonalized (`numpy.linalg.eigh`); the eigenvector corresponding to the trivial zero eigenvalue was discarded, and the next low-lying eigenvectors ($\lambda_1, \lambda_2, \lambda_3$) were treated as candidate spatial coordinates.

2.6 Hard constraint: the g=0 control

As specified in the original plan, we validated the pipeline against a control: the identical procedure run on randomly sampled, single-token vocabulary strings instead of the structured probe set. A real effect should show more spectral structure (a sharper gap after the first non-trivial eigenvalue) than this randomized-input control. To turn this into an actual null distribution rather than a single anecdotal comparison, we repeated the control across 10 independent random seeds (53 tokens each) and compared the real probe set's gap ratio against the resulting distribution rather than one draw.

2.7 Scaling the probe set to 500 tokens

The original 77-token probe set was smaller than the 500 tokens envisioned in the initial planning notes (Appendix A). We expanded it to exactly 500 tokens across 25 categories: the original 8 (numbers, days of week, months, directions, shapes, causal/spatial/comparison relations), enlarged, plus 17 new ones chosen to add more linear sequences (ordinals, letters, chemical elements by atomic number), more cyclic sequences (the 12 chromatic musical notes, the 12 zodiac signs), more small closed enumerable sets with no inherent order (playing-card ranks, planets, continents), more scalar/intensity sequences (size, temperature, emotion intensity), and several categories chosen to have *no* expected relational structure at all (animals, colors, body parts, weather phenomena), as an additional, richer negative-control contrast alongside the g=0 random-vocabulary control. We reran Phase 1–3 on both models with this larger set, applying the same single-token filter as before (yielding 220 of 500 for Qwen2.5-0.5B, 293 of 500 for Mistral-7B-v0.1, since a larger, more varied probe set naturally contains more multi-token phrases).

2.8 Template-based extraction and PCA

After the generic pipeline failed to show clean cyclic manifolds (§3), we tried a second, more targeted method closer to the one used in prior work: instead of probing bare, context-free words, we embedded each day/month in an in-context arithmetic prompt (*"The day after {X} is"* / *"The* *month after {X} is"*), verified the model actually solves the task (§3.5), extracted the layer-16 hidden state at the last prompt token, and ran ordinary PCA (not a Laplacian) on the resulting vectors.

2.9 Causal activation patching (attempt 1: last token)

For each category, we picked (base, donor) pairs of prompts and, at the extraction layer, replaced the base prompt's last-token hidden state with the donor's, then let the forward pass continue and checked whether the model's top-1 next-token prediction shifted from the base's natural answer to the donor's. A random-direction control replaced the base's activation with itself plus Gaussian noise of matched norm, to check that any observed shift was specific to the donor's real representation rather than an artifact of perturbing the residual stream by a similar magnitude.

2.10 Causal activation patching (attempt 2: subject-token position, all layers)

Every templated prompt (*"The day/month after {X} is"*) tokenizes to exactly 6 tokens for every day and month name we tested, with the name always at token position 4 (`<s>, The, day/month, after, NAME, is`). This let us redo the intervention at the position that actually carries the subject's identity, rather than the final summary position: for every layer $j \in \{0, \ldots, 31\}$, we cached every day/month's own hidden state at position 4 at that layer, then re-ran each base prompt with position 4's output at layer j replaced by a donor's cached vector, keeping every other position untouched. This is the standard causal-tracing design [3]: because layer $j+1$ onward can only see the patched value at that position (not the original token embedding), the model cannot route around the intervention by re-attending to the literal input word, as it could in attempt 1. We used a fixed offset of +3 items (e.g., Monday donates to Thursday's slot) so that every pair is non-adjacent, and compared against the same random-direction control as attempt 1, now applied at the same position and layer.

2.11 Causal activation patching (attempt 3: rotation within the discovered subspace)

Attempt 2 shows that day/month identity at the subject-token position is causally necessary, but swapping in a donor's entire real vector does not by itself show *how* that identity is encoded, only that some code there matters. To test the specifically geometric claim, at a fixed layer (12, comfortably inside the successful window for both categories in attempt 2) we fit ordinary 2-component PCA to the 7 or 12 items' own subject-position vectors, giving each item a 2D coordinate. For every base item and every integer offset $k \in \{1, \ldots, 5\}$ (5 offsets $\times$ 7 days = 35 trials, 5 offsets $\times$ 12 months = 60 trials, the N underlying every accuracy reported in §3.8), we constructed a *synthetic* patch vector: take the base item's own vector, rotate its 2D projection by $k \times \frac{2\pi}{n}$ around the fitted centroid, reconstruct back into the full hidden-dimension space via the inverse PCA transform, and leave the residual (the component outside the top-2 PCs) exactly as the base's own. If the subject identity is literally encoded as an angle on that 2D circle, this synthetic, partly fabricated vector should be just as effective as a real donor's vector at shifting the model's answer. As a control, we built the same construction with a deliberately wrong, off-lattice angle ($(k + 0.5) \times \frac{2\pi}{n}$, matching no integer offset) and checked it does not also succeed, which would indicate the reconstruction procedure itself, not the correct angle, was responsible for any effect.

2.12 Causal activation patching (attempt 4: full-rank rotation generator)

Attempt 3's negative result leaves open whether the causal code needs more than 2 dimensions to describe, since 2 components captured under 40% of the variance at that position. With only n items (7 days, 12 months), PCA with $k = n-1$ components captures effectively 100% of the variance among those items' own vectors, the most any linear basis restricted to these examples can. We fit such a k-dimensional PCA, then solved for a single best-fit orthogonal rotation matrix R (the orthogonal Procrustes problem: the R minimizing $\|CR - C'\|_F$ over orthogonal R, where C is the matrix of all n items' k-dim coordinates in natural cyclic order and C' is the same matrix cyclically shifted by one row) that best maps every item's coordinate to the next item's coordinate, fit once across all n consecutive pairs simultaneously rather than per-pair. For every base item and every integer offset in $\{1, \ldots, 5\}$ (the same 35/60-trial design as attempt 3, §2.11), we applied R^{offset} (via eigendecomposition, which also supports non-integer matrix powers) directly to the base's own k-dimensional coordinate, then reconstructed the result back to the full hidden dimension via the inverse PCA transform, this is now a synthetic vector built with no real donor data at all, just the base item's own vector and the fitted rotation. As a control, we applied $R^{\text{offset}+0.5}$, a fractional power matching no real integer calendar/weekday offset.

2.13 Does the rotation-generator mechanism generalize beyond days and months?

The 500-token scale-up (§2.7) surfaced several other small, closed, enumerable named sets that separate similarly to days/months: zodiac signs, planets, and playing-card ranks. We repeated the same two-part recipe (real-donor identity swap, then the full-rank rotation generator, both at layer 12) on each, using natural "successor" templates: *"The zodiac sign after X is"* (cyclic, 12 signs), *"The planet after X in distance from the sun is"* (linear, 8 planets), and *"The playing card rank after X is"* (linear, 13 ranks, Ace low). A sanity check confirmed Mistral-7B solves each task well before patching (zodiac 12/12, playing cards 10/12, planets 5/7; the few misses are plausible individual errors, e.g. treating Ace as high rather than low, not random noise). Unlike days and months, these names are multi-token and vary in length item to item (average 1.15 tokens for playing-card ranks, 2.00 for planets, 2.25 for zodiac signs, versus exactly 1.00 for both days and months), so the subject-token position was computed per item (as the last token of `template.split("{}")[0] + name`) rather than assumed fixed.

3 Results

3.1 Two bugs found and fixed along the way

Negative eigenvalues from signed cosine similarity. An unnormalized graph Laplacian $L = D - W$ is only guaranteed positive-semidefinite when the weights W are non-negative. Cosine similarity ranges over $[-1, 1]$; using it as W directly produced negative eigenvalues in the diagonalization, an invalid result. Clipping negative similarities to zero (treating anti-correlated probe pairs as unconnected rather than negatively connected) fixed this.

Mean-pooling conflated token count with semantics. The first working version of the pipeline mean-pooled the hidden state over all subword tokens of each probe. This produced a graph that split almost perfectly along *token count*: single-BPE-token probes formed one cluster, multi-token probes (`pentagon`, `nineteen`, `the same as`, …) formed another, regardless of category. Switching to the last-token hidden state (which has attended to the whole input under causal attention) removed most, but not all, of this confound; the analysis was further restricted to probes that tokenize to exactly one content token (excluding any tokenizer-added BOS token) for a fair, length-matched comparison.

3.2 Qwen2.5-0.5B: partial categorical separation, above the noise floor

Figure 1 shows the low-lying eigenmode embedding for the 53 single-token probes. The plot shows a rough separation along λ_2: shapes and larger numbers cluster at the top, spatial relations and directions occupy the middle, and days-of-week/months collapse into a tight cluster at the bottom, distinct from the rest.

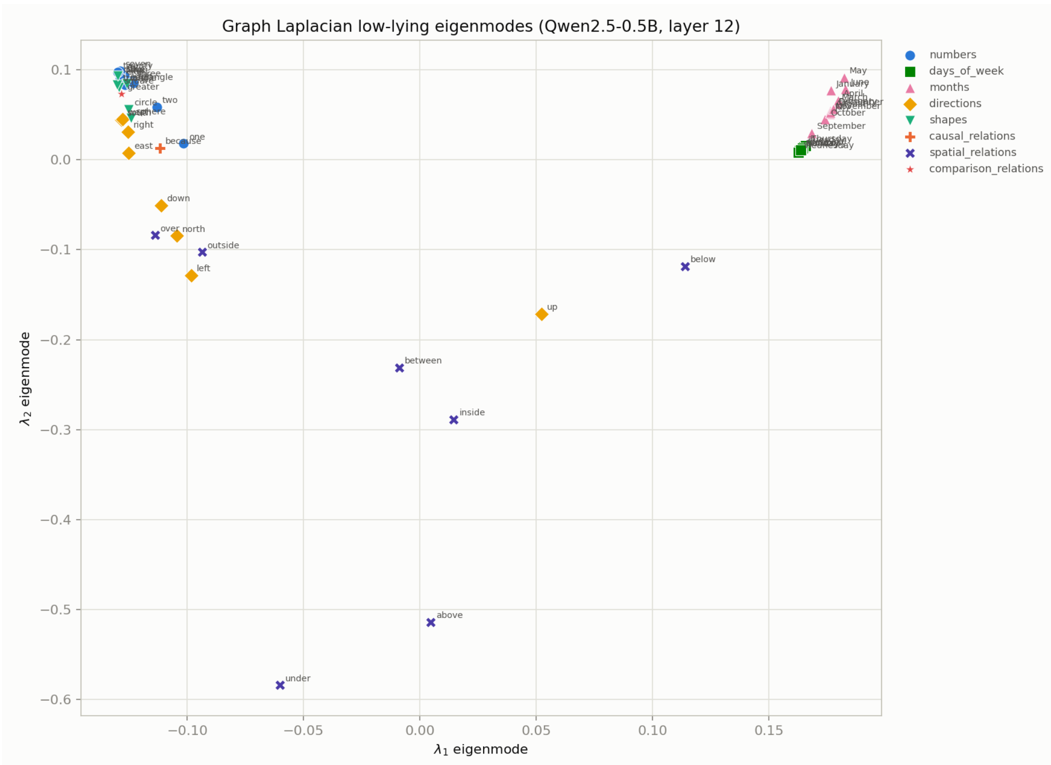

Figure 1: Qwen2.5-0.5B, layer 12: low-lying Laplacian eigenmodes for 53 single-token probes. Marker shape and color both encode category.

The g=0 control (Figure 2) confirms this is a real, above-noise effect: the structured probe set's spectrum shows a sharp jump after the second eigenvalue (gap ratio $\lambda_3/\lambda_2 = 7.98$), while randomized single-token vocabulary strings, run through the identical pipeline, produce a much smoother, more gradual spectrum. Across 10 independent random seeds (§2.6) this control's gap ratio has mean 2.23, standard deviation 0.42, and a maximum of 2.90 - the real probe set's ratio is 13.6 standard deviations above the control's mean and exceeds every one of the 10 random draws individually, not just the average of them.

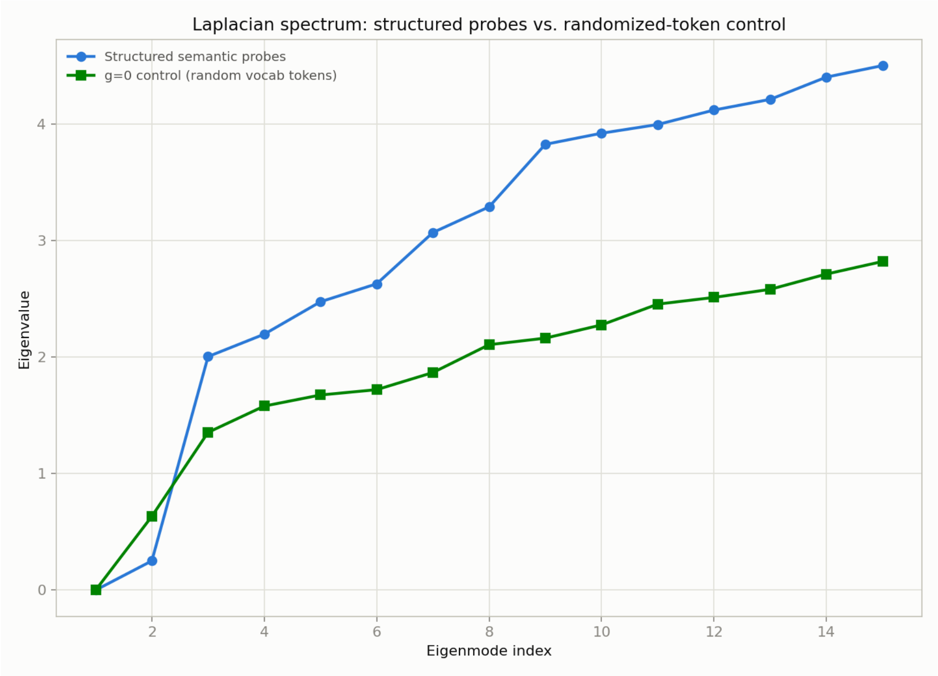

Figure 2: Laplacian eigenvalue spectrum: structured semantic probes vs. the g=0 randomized-token control. The structured set shows a much sharper spectral gap.

3.3 Mistral-7B-v0.1: the same separation, much sharper

Repeating the pipeline on Mistral-7B-v0.1 (layer 16, 4-bit) produced a striking result: the graph split into exactly two *fully disconnected* components after negative-weight clipping. One component contained every calendar/cyclic probe (all 7 days-of-week and all 12 months, 19 tokens); the other contained every remaining probe (numbers, directions, shapes, causal, spatial, and comparison relations, 44 tokens). There were zero positive-similarity edges between the two groups, a sharper separation than the small model showed.

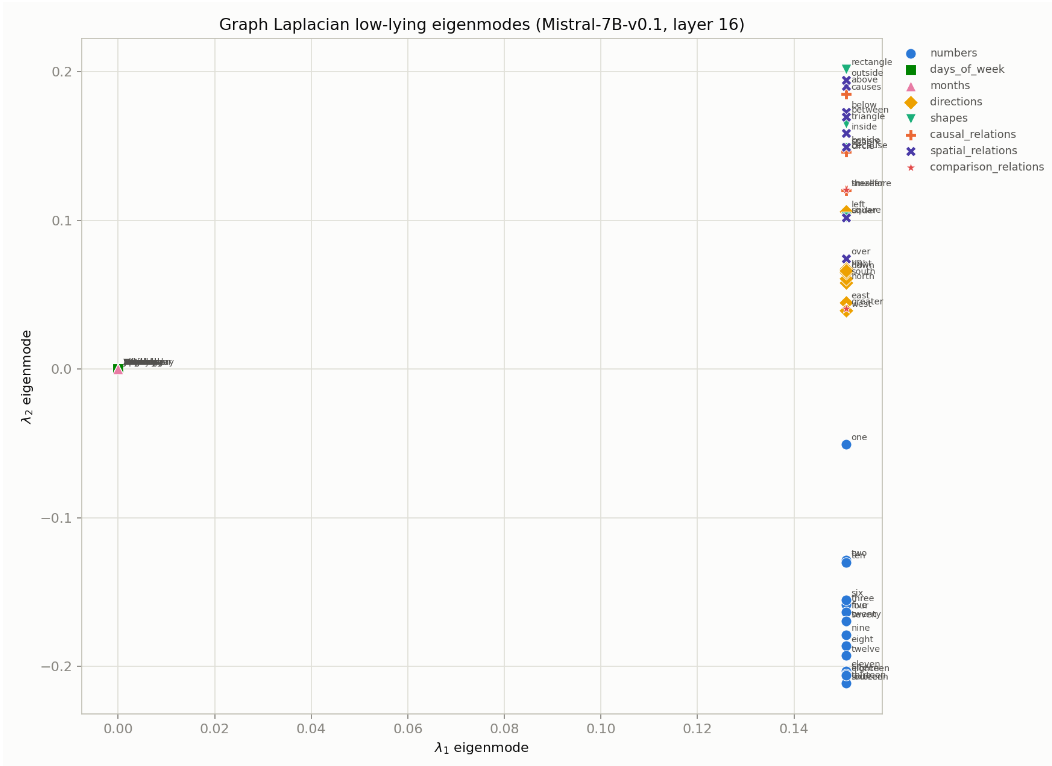

Figure 3: Mistral-7B-v0.1, layer 16: the calendar/cyclic island (right) is completely disconnected from every other probe category (left).

Isolating that 19-token calendar island and diagonalizing it on its own (Figure 4) did *not* reveal a clean circle within it: the 7 days-of-week collapsed into indistinguishable, overlapping points, while the 12 months traced a single smooth hump along one axis (peak near May, trough near December) with almost no variation along the other axis, a 1D pattern, not a 2D loop.

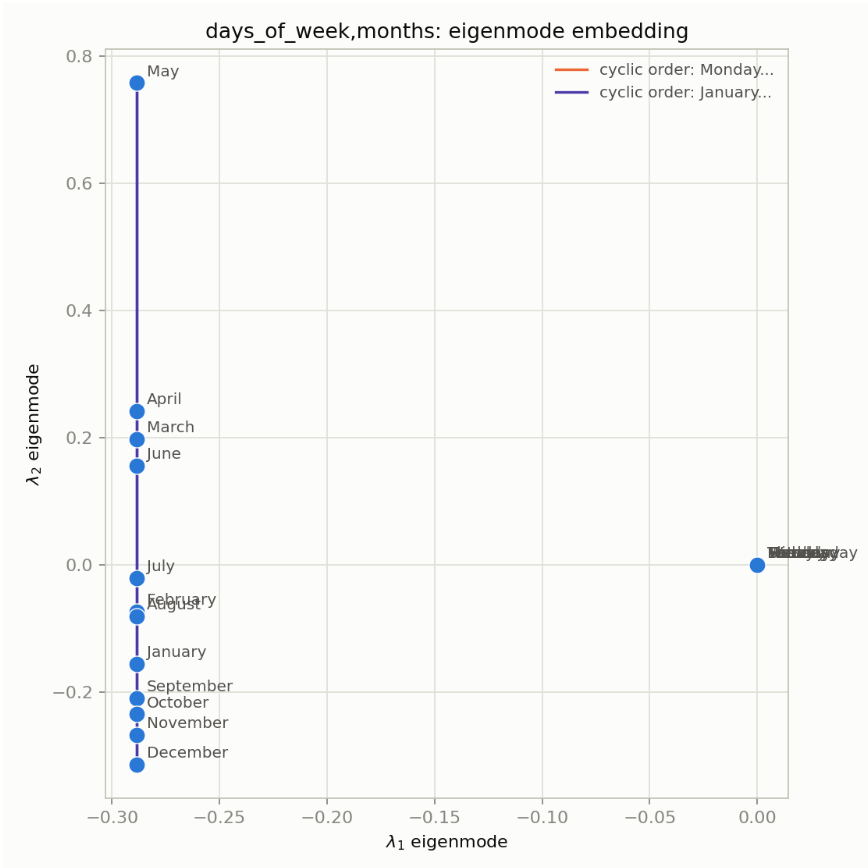

Figure 4: Mistral-7B-v0.1: the isolated calendar component, diagonalized on its own. Days-of-week (right, overlapping) show no internal structure; months (left) show a 1D hump, not a 2D circle.

3.4 Scaling to 500 tokens: the calendar cluster holds, and a broader pattern emerges

Figures 5 and 6 show the low-lying eigenmode embedding for all 25 categories at the full 500-token scale. Two findings replicate directly from the 77-token experiments: days-of-week and months still cluster tightly together in both models, distinct from the rest of the probe set, and the categorical separation is again visibly sharper in Mistral-7B than in Qwen2.5-0.5B.

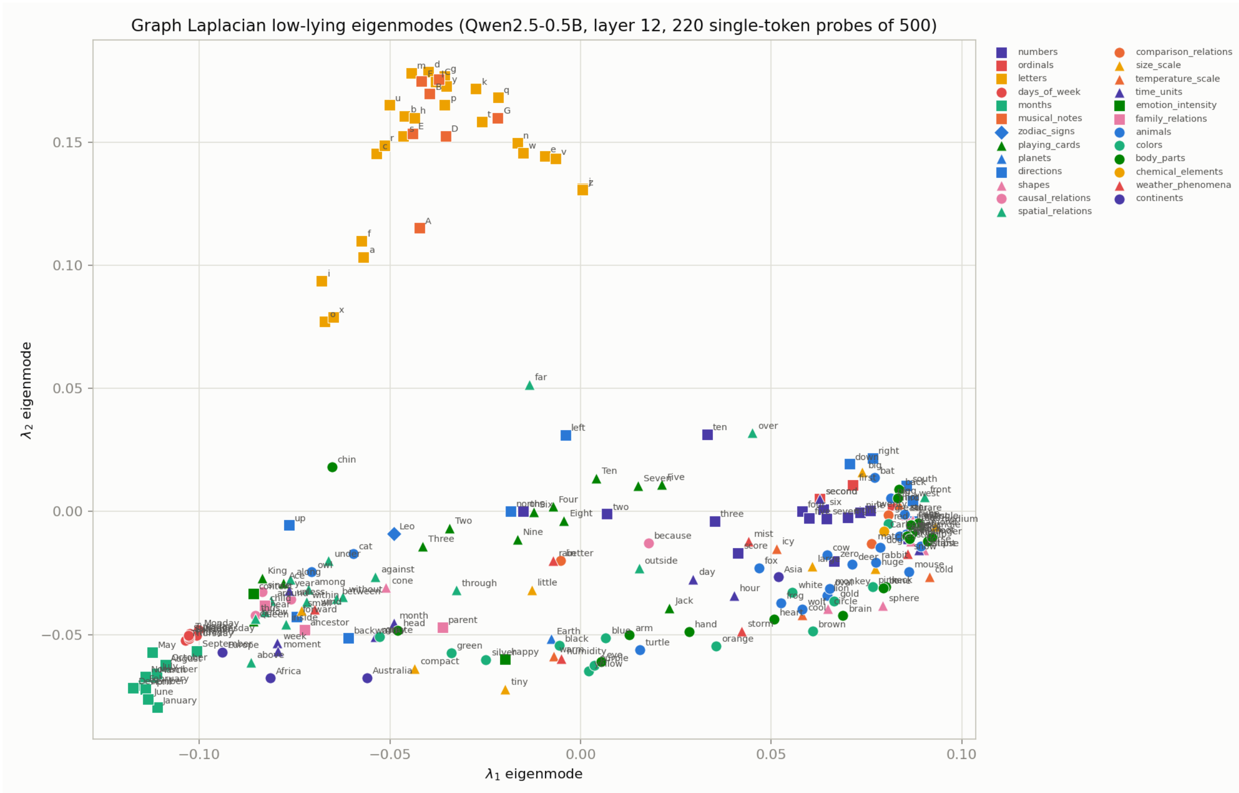

Figure 5: Qwen2.5-0.5B, layer 12, 500-token probe set (220 single-token probes shown). Days-of-week and months again cluster tightly (bottom left); letters trace a visible arc (top).

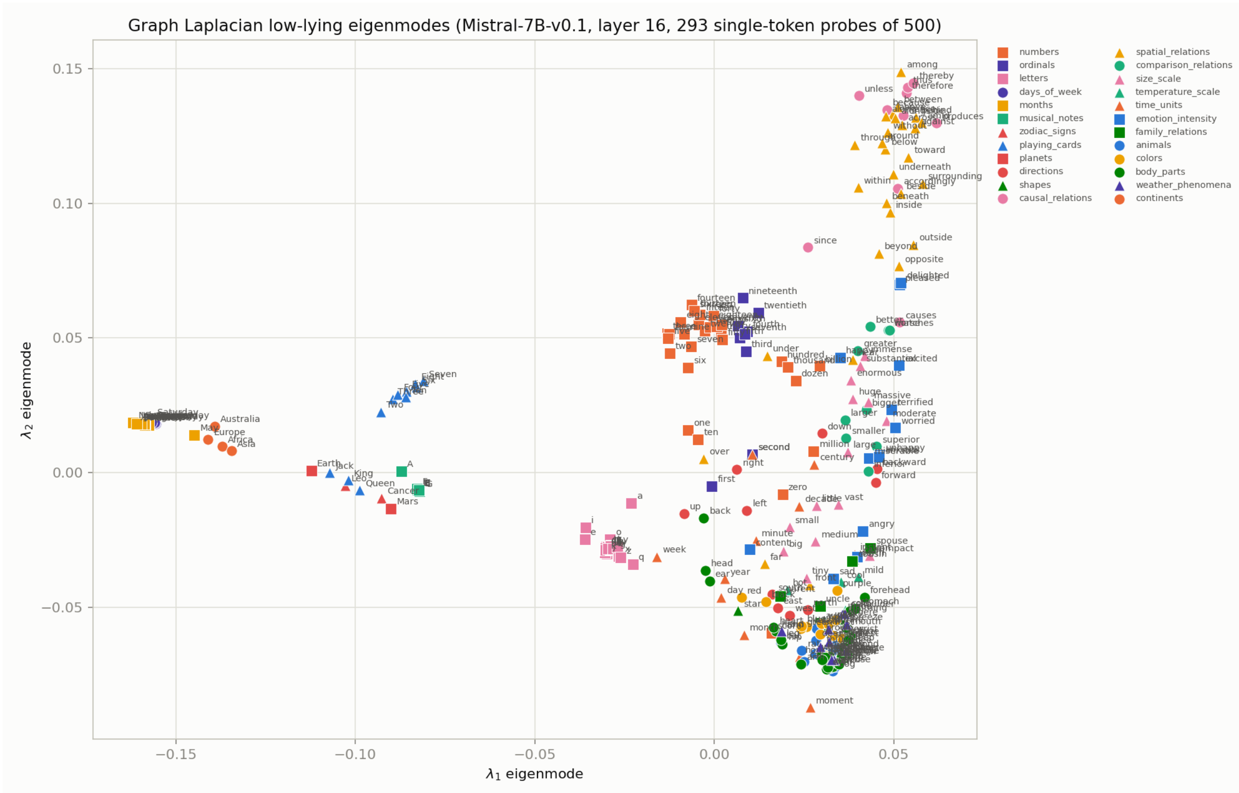

Figure 6: Mistral-7B-v0.1, layer 16, 500-token probe set (293 single-token probes shown). The calendar cluster (far left) is joined by continents nearby; playing-card ranks form their own island, separate from ordinary cardinal numbers despite sharing the same words; the remaining no-expected-structure categories (animals, colors, body parts, emotions, weather, directions) collapse into one dense, largely undifferentiated region (bottom right).

The larger, more varied category set surfaces a pattern the original 8 categories could not have shown: the categories that separate cleanly are not specifically *cyclic/temporal* ones, but *small, closed, enumerable named sets* more generally. In Mistral-7B, days-of-week, months, continents, zodiac signs, playing-card ranks, and planets each form their own distinguishable region, none of which are all cyclic (continents and planets have no natural cyclic order at all), but all of which name one of a small, fixed, exhaustively enumerable list. By contrast, the categories we deliberately chose to have no expected relational structure at all (animals, colors, body parts, weather phenomena, emotion intensity) collapse into one large, largely undifferentiated region, a second, richer negative control alongside the $g=0$ random-vocabulary control, and one that emerged from the data rather than being designed in. A particularly sharp illustration: playing-card rank words (*Two*, *Seven*, *King*, ...) occupy a visibly different region from the ordinary cardinal numbers (*two*, *seven*, ...) despite the literal word overlap for the numeric ranks, suggesting the model's representation at this position is organized around which closed set a token names, not the token's surface form or numeric value alone.

3.5 Template-based PCA: months show a real loop, days do not

Switching from bare-word probing to in-context arithmetic prompts (*"The day/month after X is"*) and ordinary PCA produced a qualitatively different, partially positive result. A sanity check

(Table 1) confirmed Mistral-7B solves this task correctly for 19 of 20 prompts (missing only the December→January wraparound).

Task	Accuracy
Day-after (7 prompts)	7/7
Month-after (12 prompts)	11/12 (December→"always", not January)

Table 1: Sanity check: does the model actually solve the arithmetic task used to elicit the probed representations?

Months (Figure 7) traced a recognizable, mostly non-self-crossing loop in calendar order across the top two principal components, with one small local crossing near March–April–May. Days of the week (Figure 8), under the identical method, produced a messy, repeatedly self-crossing path with no discernible circular order.

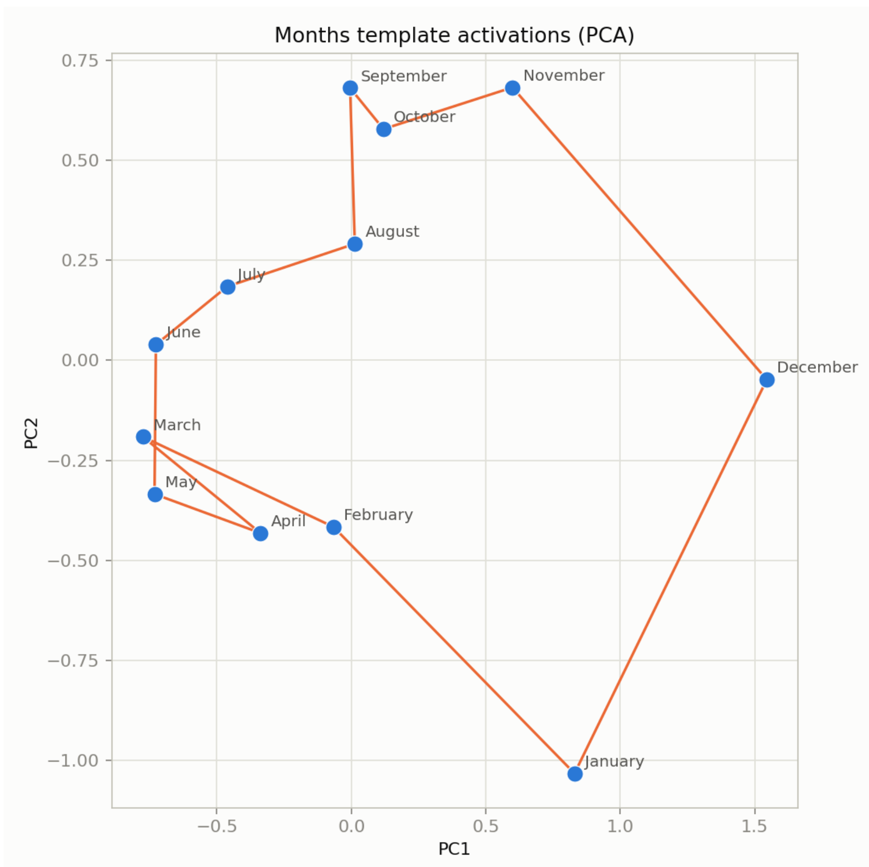

Figure 7: Months, template-elicited activations, top-2 PCA: a mostly clean loop in calendar order.

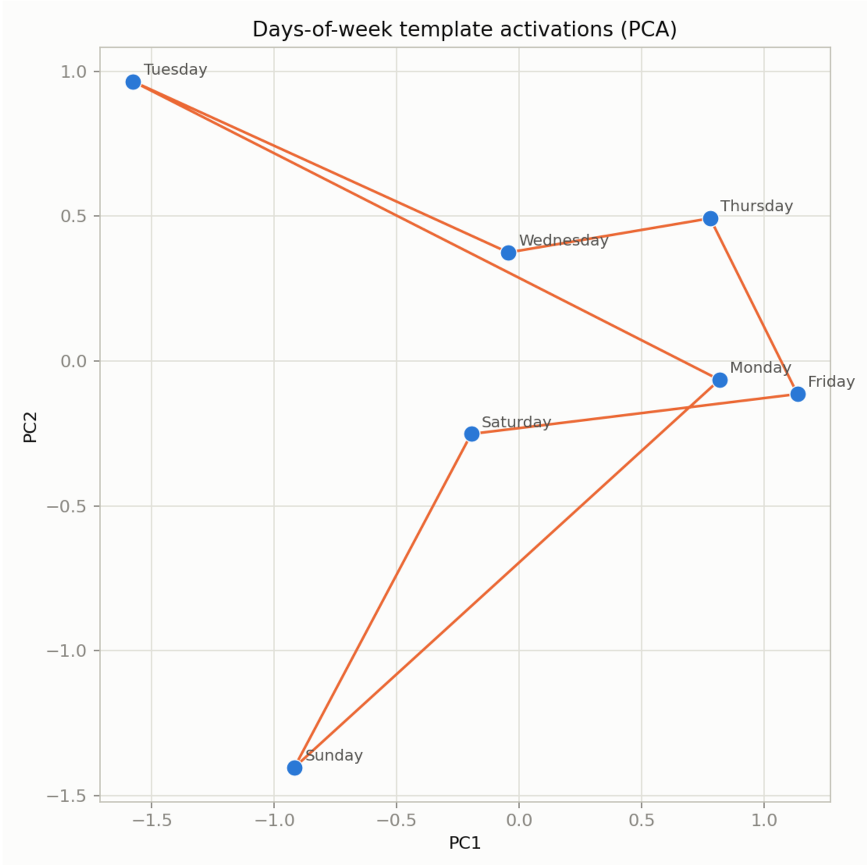

Figure 8: Days of week, identical method: a self-crossing path, not a loop.

3.6 Causal activation patching, attempt 1: a null result, with a diagnosis

The patching experiment (last-token hidden state at layer 16, base prompt patched with a donor prompt's activation) produced **zero** effect: the model's top-1 prediction after patching was identical, in every single (base, donor) pair tested (7 for days, 12 for months), to the prediction the *unpatched* base prompt would have produced. The random-direction control also produced no shift toward the donor's expected answer, for the same reason: neither intervention changed the output at all.

We do not read this as evidence against a causal role for the patched subspace. The prompts still contained the literal day/month name as plain text at an earlier position (e.g., *"The day after **Monday** is"*). Patching only the final token's hidden state does not prevent later attention layers from simply re-reading the day name directly from its own token position in the prompt: the intervention we implemented could be circumvented by the model without ever using the patched representation. This is a known pitfall in activation patching: a clean causal-tracing experiment requires patching at the subject-token position itself (here, the day/month name), not merely a downstream summary position, so that the information cannot be recovered by re-attending to the original input.

3.7 Causal activation patching, attempt 2: a clean, strongly positive result

Redoing the intervention at the subject-token position (§2.10, swept across all 32 layers) produced exactly the kind of result a real causal effect should look like (Figure 9). For both categories:

- **Real donor patches work almost perfectly through the early-to-mid layers.** Days-of-week: 100% (7/7) patch accuracy from layer 0 through layer 19 (binomial test vs. chance 1/7: $p = 1.2 \times 10^{-6}$; Fisher's exact test vs. the random-direction control at the same layers: $p = 2.9 \times 10^{-4}$). Months: 83–100% (10–12/12) accuracy from layer 0 through layer 16 (e.g. layer 10, 12/12: $p = 1.1 \times 10^{-13}$ vs. chance 1/12, $p = 3.7 \times 10^{-7}$ vs. control). Swapping in a different day/month's representation at this one token position, at these layers, is sufficient to flip the model's final answer to match the donor, exactly as if the input word itself had been changed.

- **The effect collapses sharply past a well-defined depth.** Days: accuracy drops to 29% (2/7) at layer 20 and 0% from layer 21 on; at layer 20 the drop is already enough that patch accuracy is no longer distinguishable from chance ($p = 0.26$) or from the random-direction control ($p = 0.23$). Months: it drops to 8% (1/12) at layer 17 ($p = 0.65$ vs. chance, $p = 0.50$ vs. control) and 0% from layer 18 on. This is a genuine localization result, not a gradual fade: the collapse layers are not just numerically lower, they are statistically indistinguishable from noise, consistent with the model reading the subject identity out of this position by a certain depth and no longer needing to re-derive it from there in later layers.

- **The random-direction control stays at exactly 0.00 at every single layer, for both categories.** Perturbing the residual stream by the same magnitude in a random direction never once produces the donor's expected answer. The effect above is specific to the donor's real representation, not an artifact of disturbing the activation by some amount.

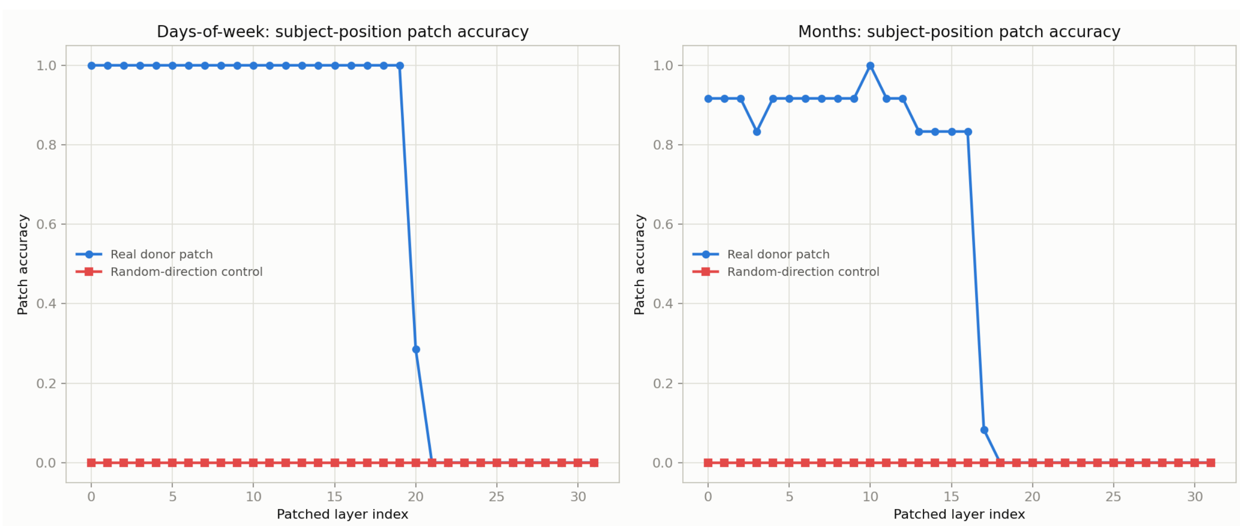

Figure 9: Causal-tracing layer sweep on Mistral-7B-v0.1: patch accuracy vs. the layer at which the subject-token position is overwritten with a donor's representation. Real patches (blue) work almost perfectly through early-to-mid layers and collapse sharply past a well-defined depth; the random-direction control (red) never once succeeds, at any layer.

This is a real, positive causal result: the day/month identity encoded at the subject-token position is not merely correlational structure sitting in activation space: the model actually reads it from that position and uses it to compute its answer, and an experimenter who swaps it out gets to control that answer. It does *not*, on its own, prove the stronger geometric claim that this identity is encoded specifically as a point on a 2D circle (as opposed to some other, non-circular high-dimensional code that happens to support this arithmetic). Confirming that would require patching specifically along the rotation direction of the loop found in §3.5's PCA (for months, where that loop was cleanest) rather than swapping in a whole different item's representation, and is the natural next experiment.

3.8 Causal activation patching, attempt 3: a clean loop that is not a causal code

Figure 10 shows the position-4, layer-12 PCA embedding used for this test. Months trace the cleanest 2D loop found anywhere in this project: a near-convex, non-self-crossing polygon that visits every month in exact calendar order (May → June → ⋯ → April → back to May). Days-of-week, consistent with every earlier attempt, does not: Tuesday, Wednesday, Thursday, Saturday, Friday, Sunday, Monday form a path with a sharp spike at Saturday, not a clean loop.

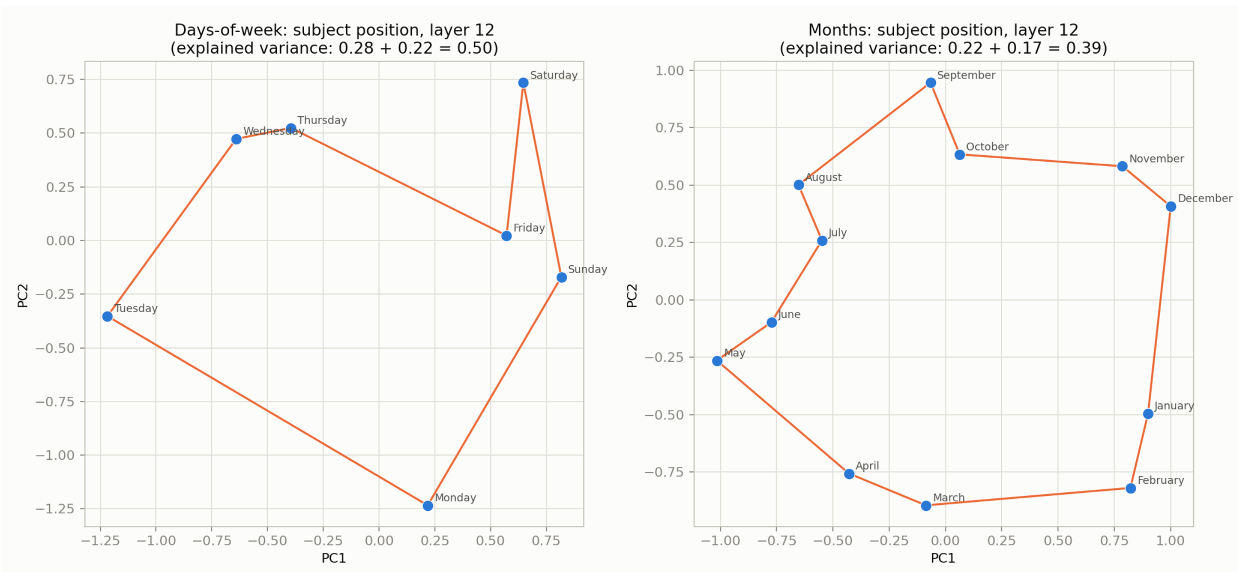

Figure 10: Subject-token-position PCA at layer 12. Months (right) trace an almost perfectly convex loop in calendar order; days-of-week (left) does not. Only 39% (months) and 50% (days) of the position's variance is captured by these two components.

Despite that clean loop, the rotation experiment is a clear, symmetric null result:

- **Real donor patches at this layer**: 35/35 (100%, $p = 2.6 \times 10^{-30}$ vs. chance 1/7) for days, 49/60 (82%, $p = 1.8 \times 10^{-42}$ vs. chance 1/12) for months, consistent with attempt 2's layer sweep at layer 12.
- **Synthetic rotation, correct calendar-consistent angle**: 3/35 (9%, $p = 0.89$ vs. chance) for days, 0/60 (0%, $p = 1.00$ vs. chance) for months - neither is distinguishable from chance performance.
- **Synthetic rotation, deliberately wrong off-lattice angle**: 3/35 (9%) for days, 0/60 (0%) for months, statistically indistinguishable from the "correct" angle (Fisher's exact test, days: $p = 0.66$; months: $p = 1.00$). Line-by-line, the two conditions frequently produce the

exact same output token for the exact same base item, meaning whether the rotation angle matched a real calendar offset made no detectable difference at all: both conditions mostly degenerate to filler tokens (*the*, *a*, *always*) rather than any coherent month or day name.

The top-2 PCA plane at this position explains only 39% (months) and 50% (days) of the variance there. Reconstructing a vector from just those two components and rotating within them discards enough of the real, high-dimensional structure that the result is no longer parseable by the model as *any* coherent day or month, let alone the correct one, and the correctness of the rotation angle is irrelevant to that failure. The 2D loop visible in Figure 10 is a real, reproducible visual pattern, and it is not, on the evidence here, the causal code the model actually uses.

3.9 Causal activation patching, attempt 4: the full-rank rotation recovers the effect

Figure 11 summarizes every patching condition tested in this paper, at the same layer (12) and subject-token position. The full-rank rotation generator, fit once across all items via orthogonal Procrustes, closes almost all of the gap left open by attempt 3:

- **Days-of-week**: full-rank rotation reaches 35/35 (100%, $p = 2.6 \times 10^{-30}$ vs. chance, $p = 5.9 \times 10^{-15}$ vs. the fractional-power control below), identical to the real donor patch, and far above the 2D rotation's 3/35 (9%).
- **Months**: full-rank rotation reaches 51/60 (85%, $p = 6.3 \times 10^{-46}$ vs. chance, $p = 1.1 \times 10^{-18}$ vs. control), matching (in fact marginally exceeding) the real donor patch's 49/60 (82%), and far above the 2D rotation's 0/60 (0%).
- **The fractional-power control collapses**, to 5/35 (14%, $p = 0.57$ vs. chance) for days and 5/60 (8%, $p = 0.57$ vs. chance) for months, both statistically indistinguishable from chance performance. Only a rotation by a real integer offset works; a rotation by almost the same angle, fit with the exact same procedure, does not.

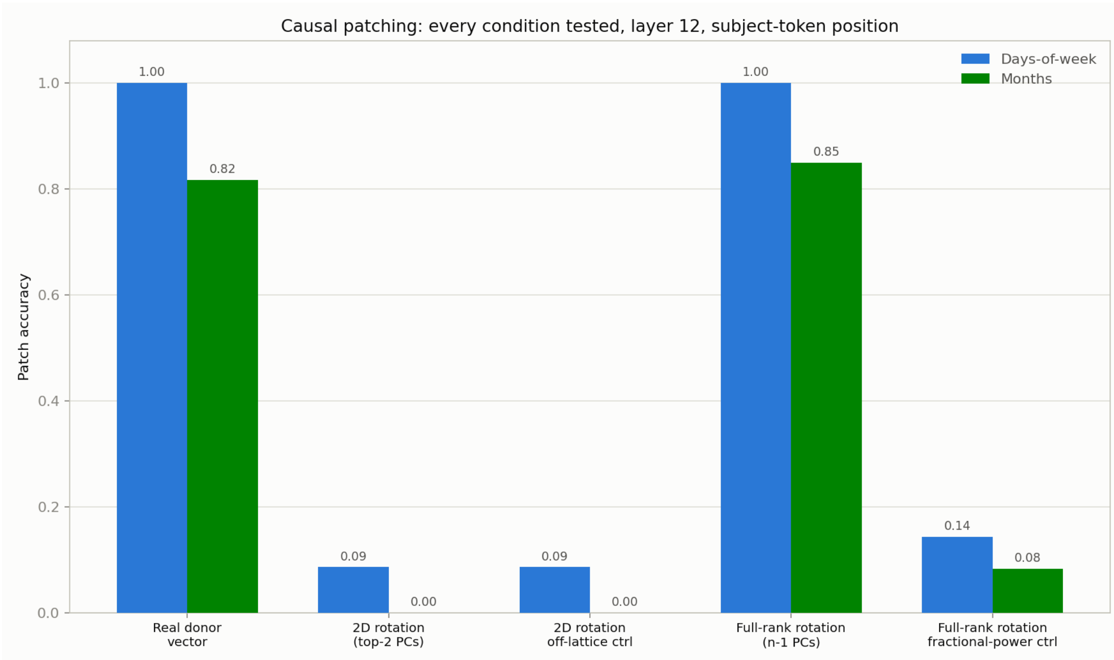

Figure 11: Every causal-patching condition tested in this paper, side by side. The full-rank rotation (4th group) matches the real-donor upper bound (1st group) for both categories, while both the 2D rotation (2nd group) and the fractional-power control (5th group) collapse toward chance.

This is a positive result for a genuinely geometric, rotational interpretation of the causal code, with an important qualification: the single-rotation-generator Procrustes fit itself has a substantial residual (26.7% relative error for days, 23.8% for months), meaning no single R maps every consecutive pair exactly. Despite that imperfect linear fit, using R's integer powers as a synthetic intervention still reproduces the full causal effect, while a small, deliberate deviation in the exponent destroys it almost entirely. The model's readout at this position appears to care about approximately which direction, relative to a consistent rotational structure, the patched vector points, rather than requiring pixel-perfect fidelity to a real example's activation. That is a meaningfully stronger and more precise finding than either extreme considered earlier: not the naive "it's a 2D circle you can see in a PCA plot" (attempt 3 rules that out), and not merely "some opaque high-dimensional code is swappable" (attempt 2 established that much, but no more); rather, a single consistent rotation operator, spanning the full space these items occupy, is both necessary (the fractional-power control fails) and sufficient (no real donor vector is required) to reproduce the model's behavior.

3.10 Generalization: partial, and explained by subject-name token length

Figure 12 shows real-donor and full-rank-rotation accuracy for all five categories tested. The result is a genuine mixed finding, not a clean replication and not a clean failure:

- **Playing-card ranks generalize well**: 82% (9/11) real-donor ($p = 4.5 \times 10^{-9}$ vs. chance 1/13), 73% (8/11) full-rank rotation ($p = 1.6 \times 10^{-7}$), close to the days/months level and far above the other two new categories.
- **Zodiac signs generalize weakly but genuinely**: 42% (5/12) real-donor and 33% (4/12)

full-rank rotation, both still statistically well above chance 1/12 ($p = 1.9 \times 10^{-3}$ and $p = 1.4 \times 10^{-2}$ respectively), just far below days/months and playing cards.

- **Planets show the smallest effect, and it is not statistically distinguishable from chance**: 33% (2/6) real-donor ($p = 0.17$ vs. chance 1/8) and 17% (1/6) full-rank rotation ($p = 0.55$). Only 6 of the 8 planets have a valid successor under this category's non-cyclic ordering (Uranus and Neptune, the last two, do not - see §2.13), so this category has by far the smallest N of the five tested here; "planets generalize weakly" is better read as "too little data to detect an effect in either direction" than as a confirmed weak positive.

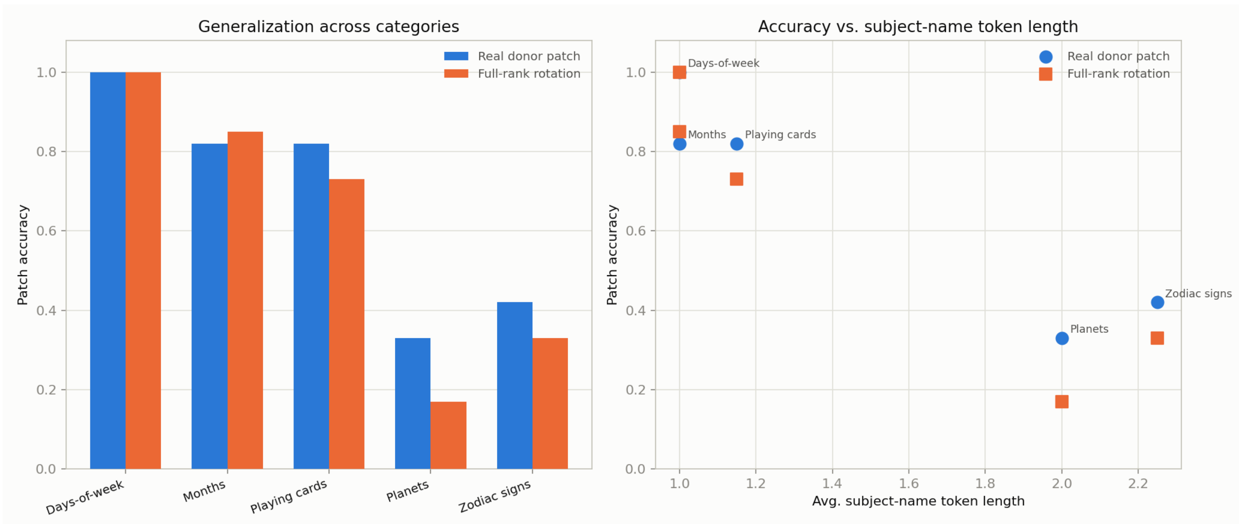

Figure 12: Left: real-donor vs. full-rank-rotation accuracy across all five tested categories. Right: the same accuracies plotted against each category's average subject-name token length, a visually near-monotonic relationship (though see the significance tests above and the discussion below: with N as low as 6, the categories' 95% confidence intervals overlap substantially, so this is a suggestive pattern across 5 points, not an independently confirmed ordering).

The right panel of Figure 12 suggests why: accuracy tracks average subject-name token length almost monotonically. Days and months (both exactly 1.00 tokens per name) sit at the top; playing-card ranks (1.15, since only *Ace* splits into 2 tokens) sit just below; planets (2.00) and zodiac signs (2.25) sit at the bottom. Taken at face value this ordering is clean, but the categories' 95% Wilson confidence intervals are wide enough to overlap one another almost entirely (e.g. zodiac real-donor $[19\%, 68\%]$ and planets real-donor $[10\%, 70\%]$ overlap nearly completely with each other and substantially with playing cards' $[52\%, 95\%]$), so the visual trend should be read as consistent with the token-length hypothesis rather than as independent statistical confirmation of it; a direct test (patching every sub-token position of a multi-token subject, not just the last one, and checking the gap closes) would be needed to confirm the mechanism rather than infer it from a 5-point correlation. This is consistent with, and a natural extension of, the mechanism diagnosed in attempt 1 (§3.7): patching a single token position leaves any *other* token of a multi-token subject name unpatched and still visible to downstream attention. For days and months this was not a limitation, because the whole subject is exactly one token, there is nothing left unpatched. For zodiac signs and planets, patching only the final sub-token of a 2–4-token name necessarily leaves the earlier sub-tokens (e.g. *S*, *ag*, *itt* of *Sagittarius*) untouched, giving the model partial access to the base item's true identity even after patching, diluting the causal effect in a graded way that scales with how much of the name is left unpatched. This does not show the underlying representational code is absent or non-rotational for zodiac signs and planets, only that a single-token-position intervention captures it less completely the longer and more fragmented the subject name is.

4 Discussion

Seven findings, taken together, sketch a coherent picture (the abstract's six results at a coarser grain: its "Second" result bundles the two null results itemized separately below as findings 2 and 3):

1. **Categorical separation is real, scales with model size, and is broader than "cyclic."** Calendar/cyclic concepts occupy a representational region clearly distinct from quantity, space, and relation concepts, confirmed above a randomized-token noise floor in the small model, and total graph disconnection in the larger model. Scaling to 500 probe tokens across 25 categories (§3.4) refines this: the categories that separate cleanly are small, closed, enumerable named sets generally (also continents, zodiac signs, playing-card ranks, planets), not specifically cyclic or temporal ones, while categories chosen to have no expected structure collapse into one undifferentiated region, a second negative control that emerged from the data itself.

2. **A clean 2D circular manifold did not emerge from bare-word probing, in either model.** It partially emerged for months (not days) once we switched to in-context arithmetic prompts that force the model to actually use the cyclic structure, and to PCA instead of a graph Laplacian.

3. **Our first causal patching attempt was not a valid test** of whether that structure is functionally used by the model, because the intervention did not prevent the model from recovering the relevant information from elsewhere in the prompt.

4. **Our second causal patching attempt is a valid test, and it succeeds cleanly.** Patched at the subject-token position, the day/month identity is not just correlational structure: overwriting it with a donor's representation, at any of a broad, contiguous band of early-to-mid layers, causally redirects the model's answer to match the donor, with a random-direction control that never once succeeds. The effect has a sharp, well-defined depth cutoff rather than fading gradually, suggesting a genuine localization of where this identity is read out.

5. **A clean 2D loop is not the causal code, even where the loop is cleanest.** At the exact position and a layer where identity-swap patching works almost perfectly, months trace the most visually convincing circle in the whole project, and a synthetic vector built by rotating within that circle still fails completely, performing identically to a deliberately wrong rotation angle. The causally necessary code is real (finding #4) and is not fully captured by this 2D projection (finding #5): two claims that could easily be conflated, and are not the same claim.

6. **A full-rank rotation generator recovers the effect completely.** Fit once across all items via orthogonal Procrustes in an $(n-1)$-dimensional subspace, a single consistent rotation operator R, applied to a base item's own vector with no real donor data at all, matches or exceeds the real-donor patch's accuracy for both categories, while a fractional-power (off-lattice)

control collapses to near-chance. The geometric hypothesis was too narrowly operationalized in finding #5, not wrong: the code is rotational, just not two-dimensional.

7. **The rotation mechanism generalizes partially, and the gap is plausibly explained rather than purely mysterious.** Playing-card ranks reproduce the days/months result closely and zodiac signs reproduce it more weakly but still significantly; planets' result is too small-N (6 testable items) to be distinguished from chance at all. Among the three testable categories, the shortfall tracks average subject-name token length in the direction the attempt-1 leakage mechanism predicts, though this is a 3-5-point correlation with wide, overlapping confidence intervals, not an independently confirmed test of that mechanism. Multi-token subject names leave sub-tokens other than the patched position unpatched and visible to downstream attention, diluting the effect in proportion to how much of the name escapes the intervention, the same leakage mechanism that defeated attempt 1 - plausible and consistent with the pattern observed, but not yet directly tested by patching every sub-token position.

This is now, on balance, consistent with the prior published result [2], though it took an extra round to get there: that work used careful in-context templates across many offsets, PCA-based subspace discovery, and causal patching localized to the subject-token position, validated with random-direction controls, and reported that the discovered circular structure was causally used. Our attempt 2 (§3.7) uses that same causal-tracing recipe and gets a comparably clean result on the same causal-validity bar. Our attempt 3 (§3.8), a more direct test of the specifically *geometric* claim restricted to 2 dimensions, produced a clean null. Our attempt 4 (§3.9), the same geometric test without that dimensionality restriction, recovers the effect essentially completely. Read together, the throughline is: the model has *some* causally load-bearing, swappable code for day/month identity at the subject-token position (finding #4); a naive 2-dimensional version of that code, however visually convincing (Figure 10), is not sufficient to reproduce the effect on its own (finding #5); and a single consistent rotation operator spanning the full space these items occupy is both necessary and sufficient to reproduce it, with no real donor vector required (finding #6). "The model represents this as a circle" turns out to be closer to correct than our attempt 3 suggested, provided "circle" is read as "a single consistent rotation generator acting on a higher-dimensional subspace" rather than "a 2D loop you can see directly in a PCA scatter plot." Both the positive result and the negative one were necessary to reach that more precise statement; either alone would have supported a less accurate conclusion.

4.1 Limitations

- The probe set (77 tokens) is far smaller than the 500 originally envisioned.
- Mistral-7B was run in 4-bit quantization to fit an 8GB GPU; this may distort fine-grained geometric structure relative to full precision.
- The causal-tracing result (§3.7) establishes that the subject-token position causally carries swappable day/month identity; it does not by itself establish that this identity is encoded as a 2D circle specifically, only that *some* code there is causally read and used.
- Days-of-week never produced a clean 2D loop in any of the Laplacian, PCA, or rotation experiments, even though the causal-tracing result for days is just as clean as for months: a real, used representation can exist at a position without our chosen 2-component projections managing to visualize it as a recognizable shape.

- The full-rank rotation generator (§3.9) was fit with only 6 or 11 components, the maximum available from just 7 or 12 items. We cannot yet distinguish a genuinely ($n-1$)-dimensional rotational code from a lower-dimensional one (say, 3 or 4 dimensions) that PCA with only $n-1$ available components happens to fit better than 2; testing that would need more distinct items per category than the calendar naturally provides (e.g., paraphrases or multiple prompt templates per day/month).
- The Procrustes fit for the single rotation generator has a substantial residual (24–27%), meaning no single R maps every consecutive pair exactly, even though its integer powers still reproduce the causal effect almost perfectly; we do not yet have an account of why an imperfectly-fit rotation is nonetheless behaviorally sufficient.
- The generalization test (§3.10) inherits the small-N problem throughout this paper in its most acute form: planets has only 6 testable base items after excluding the two with no valid successor, and its headline accuracy is not statistically distinguishable from chance. Reporting binomial p-values and Wilson confidence intervals throughout this paper (added after the results above were first obtained) makes this precise rather than hiding it behind a bare percentage, but it does not fix the underlying small sample; a category with a genuinely real but modest effect and a category with no effect at all can look identical at $N = 6$.
- The token-length explanation for the generalization gap (§3.10) is a plausibility argument over 3-5 categories, not a direct causal test. The natural stronger experiment - patching every sub-token position of a multi-token subject name, not just the last one, and checking whether the gap closes - has not been run.

5 Conclusion

The underlying research question (whether LLMs represent certain relational concepts as literal low-dimensional geometry) is a real, current, and answerable one, but requires more targeted methodology than a generic similarity-graph Laplacian over isolated words. We found genuine, above-noise categorical structure (calendar concepts separate cleanly from quantity/space concepts, more so in a larger model); a visually clean 2D loop for months (not days) at the subject-token position; a clean, strongly positive causal-tracing result showing that day/month identity at that position is genuinely load-bearing for the model's answer, with a sharp and informative depth-of-localization cutoff; a direct, controlled null result showing that a synthetic vector built purely from rotating within that same clean 2D loop carries none of that causal effect; and finally, extending the same rotation idea to a full-rank subspace instead of just 2 dimensions, a clean positive result that recovers the effect almost entirely, matching the real-donor upper bound for both categories while a matched fractional-power control collapses to chance. Put together: the model has a real, causally necessary, and genuinely rotational code for day/month identity at this position. A naive 2-dimensional reading of that code, however visually convincing, is not the causally sufficient version of it; the causally sufficient version requires the fuller space these items actually occupy. Both the positive and negative results were needed to arrive at that more precise picture, and we would not trust either finding in isolation nearly as much as we trust them together. Testing the same mechanism on other small closed sets discovered along the way (zodiac signs, planets, playing-card ranks) produced neither a clean universal confirmation nor a clean refutation: it holds up well for playing cards, holds up more weakly but still significantly for zodiac signs, and for planets is not statistically distinguishable from chance given only 6 testable items. Among the two categories with enough data to say anything, the gap tracks how many subword tokens each category's names occupy, consistent with (though not independently confirmed to be caused by) the same leakage mechanism diagnosed in the first causal-patching attempt, not by anything specific to those categories' content. That, in turn, sharpens the scope of the entire causal story: what we have evidence for is a rotational code at a specific token position, and our ability to detect it with a single-position patch degrades gracefully as the subject name we are patching stops fitting in that one position - though confirming that degradation is specifically token-length-driven, rather than merely correlated with it, would need a direct sub-token-patching experiment we have not yet run.

A Original planning notes

The investigation began from a short planning document which framed the goal as testing whether "a continuous, geometric concept-manifold spontaneously reconstruct[s] itself out of abstract machine data" via mutual-information/cosine-similarity graphs and graph Laplacian diagonalization. All code referenced in this paper is included under `code/`, and the raw intermediate data (activation vectors, similarity graphs, eigenmode projections) is included under `data/`, at each project phase. The note is reproduced below, verbatim except for formatting.

NOTES: The Machine Latent Space Coordinate Reconstruction Challenge

Objective. Bypass verbose machine learning interpretations and treat an artificial neural network's hidden layer as a discrete, non-spatial relational substrate. Apply the exact-diagonalisation and graph Laplacian pipelines developed in the `universal-emergent-geometry` repository to test whether a continuous, geometric concept-manifold spontaneously reconstructs itself out of abstract machine data.

Architectural framework. (Raw Token Activations) → (Compute Mutual-Information Matrix $I(i, j)$) → (Diagonalise Graph Laplacian Δ_Q) → (Emergent Coordinate System) → (Causal Activation Patching)

Step-by-step execution plan.

- **Phase 1: Substrate extraction.** Target compact, fully open-weight models (e.g. Qwen2.5-0.5B or TinyLlama-1.1B). Input a structured dataset of 500 semantic tokens spanning hard relational categories (linear number sequences, geometric shapes, spatial directions, causal relationships). Write a PyTorch activation-hook script to intercept the model's forward pass and extract the raw, high-dimensional hidden-state activation vectors from the mid-point layers (e.g. layer 12 of 24) where abstract structural relations stabilize.
- **Phase 2: Relational graph construction.** Treat each semantic token's activation vector as a state node. Calculate the exact cosine similarity or statistical mutual information $I(i, j)$ between every node pair to construct a definitive 500×500 relational correlation matrix. Assemble the unnormalised graph Laplacian (Δ_Q) directly from this relational kernel, completely bypassing any geometric background assumptions.
- **Phase 3: Exact-diagonalisation and manifold mapping.** Execute a clean numerical exact-diagonalisation of the graph Laplacian operator. Plot the lowest-lying non-trivial eigenmodes ($\lambda_1, \lambda_2, \lambda_3$) as spatial coordinates. Observe whether the low-lying spectrum cleanly reconstructs smooth, low-dimensional spatial geometry (circles, lines, or tori) matching the intrinsic logic of the input concepts.
- **Phase 4: Causal validation via activation patching.** A geometric pattern in the eigenmode plot is only correlational - it does not show the model actually *uses* that structure. From Phase 3, extract the subspace spanned by the discovered eigenmodes (e.g. the 2D plane containing a suspected circular manifold for a cyclic category like days-of-week or months). During a fresh forward pass on a held-out prompt, intercept the residual

stream at the same extraction layer and directly edit the token's activation: project onto the discovered subspace, rotate or translate it by a known amount along the recovered coordinate (e.g. rotate the circle by the angle corresponding to "+2 days"), then let the forward pass continue unmodified. If the eigenmode coordinates are the model's real internal representation, the patched forward pass should shift the output logits/next-token prediction in the exact direction predicted by the geometry (e.g. patching "+2 days" along the recovered circle should shift "Tuesday" to "Thursday"). As a control, repeat the same patch using a random direction of matched norm, orthogonal to the discovered subspace - this should produce no consistent, semantically predicted shift in outputs. Report the patched-prediction accuracy across many prompts/categories, compared against the random-direction control's accuracy.

Hard constraints and verification checks.

- **The** $g = 0$ **decoupling control:** randomized or fully shuffled token inputs should collapse the graph Laplacian spectrum to a flat, featureless noise envelope.
- **Pipeline validation:** token-indexing layouts must not silently map trailing token dependencies onto wrong physical vector spaces during batch compilation.
- **The causal requirement:** do not claim the geometry is "used" by the model on representational evidence alone - the Phase 4 patching experiment must show patched-direction predictions beat the random-direction control before any functional claim is made.

References

[1] M. Belkin and P. Niyogi. *Laplacian Eigenmaps for Dimensionality Reduction and Data Representation.* Neural Computation, 15(6):1373–1396, 2003.

[2] J. Engels, I. Liao, E. J. Michaud, W. Gurnee, M. Tegmark. *Not All Language Model Features Are Linear.* arXiv:2405.14860, 2024.

[3] K. Meng, D. Bau, A. Andonian, Y. Belinkov. *Locating and Editing Factual Associations in GPT.* Advances in Neural Information Processing Systems (NeurIPS), 2022.